Sriram Bharadwaj

Department of Physics, UCLA sriramsb@ucla.edu Website Google Scholar & InspireHEP

Education

University of California, Los Angeles *Sept 2021 – Present*
PhD in Theoretical Physics

- **PhD advisors**: Eric D'Hoker and Leo Zhou, **Mentors**: Di Luo and Zhong-Bo Kang
- **Awarded**
  1. "UCLA Physics & Astronomy Graduate Research Award" (2026-27), amount: $30k
  2. "UCLA Robert Finkelstein Fellowship in Theoretical Physics" (2022), amount: $6k
  3. " Mani L. Bhaumik Graduate Fellowship in Theoretical and Quantum Physics" (2023-26), amount: $6k/summer
- **Semi-finalist** (top 20 of 133) in the **XPrize Quantum Applications** competition (2025): "CV-QAOA: Efficient Low-Depth Quantum Optimization for Continuous Variables" – with Di Luo and Leo Zhou
- **Scientific computing skills**: tensor networks/DMRG using Julia and iTensor, exact diagonalization, variational quantum algorithms, numerical optimization, neural quantum states, Python/SciPy/PyTorch etc.

University of Oxford *Oct 2020 – Sept 2021*
MSc in Mathematical and Theoretical Physics

- Graduated with distinction and completed a double-unit master's dissertation on string amplitudes

University of California, Los Angeles *Sept 2017 – Sept 2020*
BS in Mathematics and Applied Science

- Graduated *summa cum laude* and was on Dean's honors list every quarter

Publications (in reverse chronological order)

1. **CV-QAOA: Efficient Low-Depth Quantum Optimization of Continuous Variables** *Oct 2026*
   with Di Luo and Leo Zhou, arXiv:2610.06815, submitted to *Quantum Information Processing 2027* conference

2. **Confinement Versus Screening in the Schwinger Model on AdS_2 from Bosonization and Tensor Networks** *Jul 2026*
   with Zhong-Bo Kang, Jack Isen, arXiv:2607.19468, accepted *Journal of High-Energy Physics*

3. **Information-Theoretic Scaling Laws of Neural Quantum States** *Mar 2026*
   with Y. Lu, D. Rathore, and Di Luo, arXiv:2603.23468, submitted to *Physical Review Letter* and *Quantum Information Processing 2027* conference

4. **Hamiltonian Lattice QED_3 with One and Two Flavors of Wilson Fermions: Topological Structure and Response** *Mar 2026*
   with Di Luo, Lena Funcke, Karl Jansen, et al., accepted *Physical Review D*; arXiv:2603.05616

5. **Path to Quantum Simulations of Topological Phases: (2+1)D Quantum Electrodynamics with Wilson Fermions** *Apr 2025*
   with Di Luo, Lena Funcke, Karl Jansen, et al., accepted *Physical Review Letters*; arXiv:2504.21828

6. **Non-Invertible Defects on the Worldsheet** *Sep 2024*
   with P. Niro and K. Roumpedakis, published in *JHEP* 03 (2025) 164

7. **Approaching Argyres-Douglas Theories** *Oct 2023*
   with Eric D'Hoker, published in *JHEP* 06 (2024) 082

Selected ongoing research (in order of planned submission)

Exponentially Accurate Adiabatic State Preparation for Lattice Gauge Theories and Bosonic Systems *with Di Luo*

- Proves adiabatic theorem for truncated unbounded Hamiltonians with errors exponentially small in runtime uniformly in the cutoff; applied to $(2+1)$D U(1) and SU(2) lattice gauge theories and the Hubbard-Holstein model
- Improvemed scaling with cutoff over previously considered algorithms for $(2+1)$D lattice gauge theories, superpolynomial improvement over state-of-the-art classical methods for the Hubbard-Holstein model

Simulating Topological Phases of QED_3 on a Quantum Computer Using VQE *with Di Luo et al.*

- VQE to prepare topological ground states of QED_3 with rigorous gate-complexity estimates
- Ongoing experimental validation of the predicted topological phase diagram of 2504.21828 and 2603.05616 on superconducting quantum hardware

List of presentations (in reverse chronological order)

- **HEP-QIS Seminar:** Lawrence Berkeley National Laboratory, Host: Christian Bauer (Oct 2026)
- **Rising Researchers Seminar**: Institute for Nuclear Theory, UW Seattle, Host: Yukari Yamauchi Link (Oct 2026)
- **Talk:** "Quantum Simulations of Topological Phases: (2+1)D Lattice QED with Wilson Fermions", University of Illinois Urbana-Champaign, Host: Patrick Draper (Aug 2026)
- **High-Energy Theory Seminar:** "Quantum Simulations of Topological Phases", Argonne National Laboratory, Host: Kyle Lee (Aug 2026)
- **Nuclear Theory/QIS Seminar:** Center for Frontiers in Nuclear Science, Stony Brook University, Host: Felix Ringer (Aug 2026) Link
- **Seminar:** "Quantum Simulations of Topological Phases: (2+1)D Lattice QED with Wilson Fermions", Boston College, Host: Xiao Chen (Aug 2026)
- **Contributed talk**: "Quantum Simulations of Topological Phases: (2+1)D Lattice QED with Wilson Fermions" (arXiv: 2603.05616), IAIFI Workshop, MIT (Aug 2026). Link
- **Contributed talk**: "Quantum Simulations of Topological Phases: (2+1)D Lattice QED with Wilson Fermions" (arXiv: 2603.05616), The 43rd International Symposium on Lattice Field Theory (Lattice 2026), University of Maryland, College Park (Jul 2026). Link
- **Poster**: "Quantum Simulations of Topological Phases in (2+1)D Lattice Gauge Theory" (arXiv: 2504.21828), "QuantHEP 2025", Lawrence Berkeley National Laboratory (Oct 2025). Link
- **Lightning talk**: "Quantum Simulations of Topological Phases in (2+1)D Lattice Gauge Theory" (arXiv: 2504.21828), NSF Institute for Artificial Intelligence and Fundamental Interactions (IAIFI) Summer School, Harvard (Aug 2025). Link
- **Contributed talk**: "Topological Phases and Quantum Hamiltonian Simulations of Wilson Fermions Coupled to U(1) Gauge Fields", "Probing the frontiers of nuclear physics with AI at the EIC (II)", Stony Brook University (Mar 2025). Link
- **Gong-show talk**: "Non-invertible symmetries on the worldsheet" based on *JHEP* 03 (2025) 164, "2024 IHES Summer School - Symmetries and Anomalies: a Modern Take", IHES, France (June 2024). Link
- Pedagogical presentations in the High-Energy Theory Journal Club at UCLA.
  1. Confinement vs. screening in QED_2 on AdS_2 (2026)
  2. Anomalies and instantons in gauge theory (2024)
  3. Liouville field theory and 2D quantum gravity (2023)
  4. Quantum information theory and entanglement entropy in gravity (2023)
  5. Aspects of confining strings in gauge theory (2022)

Teaching and Professional Experience

- Refereed a paper submitted to the Quantum Information Processing (QIP) conference, 2025
- Was a teaching assistant for 10+ undergraduate courses, both elementary and advanced
- Appointed as lead teaching assistant to manage and update a course with over 900 students

List of conferences and schools (in reverse chronological order)

- The 43rd International Symposium on Lattice Field Theory (Lattice 2026), UMD, College Park, 2026
- Institute for AI and Fundamental Interactions Workshop 2026, MIT
- Institute for AI and Fundamental Interactions Summer School 2026, University of Massachusetts, Boston
- Quantum Winter School 2026: Quantum Simulation at Institute for Pure and Applied Mathematics, UCLA
- QuantHEP 2025, Lawrence Berkeley National Laboratory
- Institute for AI and Fundamental Interactions Summer School and Workshop 2025, Harvard University
- 2024 Summer School on Symmetries and Anomalies, IHES, France
- 50 years of QCD, UCLA

References

- Leo Zhou, Assistant Professor of Electrical and Computer Engineering, UCLA, ✉ leoxzhou@ucla.edu
- Di Luo, Associate Professor of Physics, Tsinghua University (formerly UCLA ECE), ✉ diluo@tsinghua.edu.cn
- Zhong-Bo Kang, Professor of Physics, UCLA, ✉ zkang@physics.ucla.edu
- Eric D'Hoker, Distinguished Research Professor in Theoretical Physics, UCLA, ✉ dhoker@physics.ucla.edu